Reg. No. | | | | | | | | | | | [14CE63]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

WASTE WATER ENGINEERING

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions selecting atleast **TWO** from each part.
2. Missing data may be suitably assumed.

PART - A

1. a. Discuss the types of Sewerage systems and their Suitability. **06**

   b. Calculate the Design waste water discharge by Rational Formula for the following data: **08**

   Population of town=30000
   Average coefficient of Runoff = 0.60
   Time of Concentration of Design rain = 30min
   Waste water flow = 80% of water Supply
   Per Capita Water Demand = 120 Lpcd

2. a. Discuss the term Hydraulic Characteristics of Circular Sewer Sections running Full or Partially full. **06**

   b. Calculate the diameter and Discharge of a Circular Sewers laid at a slope of 1in 400 When it is running half- Full with a velocity of 1.9 m/sec. Take Manning's Constant n=0.012. **08**

3. a. List and explain the various Sewer Appurtenances. **08**

   b. Explain the term Maintenance and Cleaning of Sewers. **06**

4. a. Distinguish between BOD and COD. **06**

   b. The BOD of a sewage incubated for one day at 30^0C has been found to be 110mg/l. Calculate the 5-day 20^0C BOD? Take K_D= 0.1 at 20^0C. **08**

   c.

PART B

5. a. Draw a neat Flow diagram for a treatment of Domestic Sewage from a Major City. And explain the Unit operations and Unit Processes. **08**

   b. Design a Circular primary settling tank for following data. **06**

   Sewage flow = 12MLD
   Detention Time = 2Hrs
   Surface loading = 40000 L/m^2/day.

6. a. With a neat sketch explain the construction and operation of Trickling filter. **08**

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	b.	Explain the advantages and Disadvantages of Activated Sludge Plant.	**06**
7.	**a.**	Explain the stages in Sludge Digestion Process.	**07**
	b.	Explain the waste water recycling and reuse in buildings.	**07**
8.	**a.**	Explain in brief the Disposal of Waste water in Sea.	**05**
	b.	Explain the following **i.** Sewage farming. **ii.** sewage sickness. **iii.** Oxygen Sag Curve.	**09**